

SANDERSON, TX, USA

WMO: 747300

Lat: 30.167N

Lon: 102.417W

Elev: 865

StdP: 91.36

Time Zone: -6.00 (NAC)

Period: 82-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.5	-2.5	-14.1	1.2	2.9	-11.7	1.5	5.2	10.1	9.1	9.0	8.7	2.3	0	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	37.0	19.8	35.7	20.1	34.5	20.2	23.2	30.7	22.7	30.4	22.1	30.2	3.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	17.6	26.1	20.5	16.9	25.7	19.9	16.2	25.4	73.6	30.3	71.3	30.7	69.4	30.4	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	6.9	5.7	DB	-8.2	40.4	4.3	1.6	-11.3	41.5	-13.8	42.4	-16.3	43.3	-19.4	44.5	(4)
(5)			WB	-9.1	24.1	3.6	1.0	-11.7	24.9	-13.7	25.4	-15.7	26.0	-18.3	26.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.8	8.7	11.4	15.0	19.7	24.1	27.0	27.4	27.3	24.0	18.8	13.2	9.2		
	DBStd	7.84	4.21	4.83	4.82	3.96	3.20	2.38	1.96	1.90	3.24	3.91	4.71	4.62		
	HDD10.0	221	75	36	14	1	0	0	0	0	0	2	23	70		
	HDD18.3	1144	300	198	122	31	2	0	0	0	4	42	161	283		
	CDD10.0	3451	33	74	169	293	436	509	539	538	421	274	119	46		
	CDD18.3	1332	0	3	19	72	179	259	281	279	175	56	7	1		
	CDH23.3	15560	24	103	316	998	1961	3001	3377	3233	1732	650	134	31		
	CDH26.7	7249	3	32	102	412	942	1522	1715	1588	698	207	27	3		
(14) Wind	WSAvg	2.7	2.6	2.7	3.2	3.1	2.9	2.8	2.6	2.6	2.5	2.3	2.4	2.5		
(15) Precipitation	PrecAvg	359	13	18	20	17	38	48	54	44	43	36	19	9		
	PrecMax	501	42	54	87	47	138	133	271	120	120	129	95	51		
	PrecMin	222	0	3	0	1	2	0	1	3	3	0	0	0		
	PrecStd	86	14	13	25	14	37	40	64	34	38	33	24	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.1	31.1	32.2	35.5	38.1	39.9	39.0	36.7	35.6	33.0	29.4	26.7		
		MCWB	11.5	13.8	14.6	16.0	18.6	20.6	21.1	20.6	20.4	17.0	15.1	12.5		
	2%	DB	23.2	26.3	29.7	33.0	36.2	36.8	36.5	35.4	33.7	31.0	26.5	23.2		
		MCWB	10.9	12.6	13.9	15.5	18.4	20.2	20.8	20.9	20.1	17.5	14.4	11.8		
	5%	DB	20.6	23.4	26.9	31.2	34.0	35.1	34.9	34.4	32.1	29.0	24.7	20.3		
		MCWB	9.8	11.3	13.5	15.1	18.4	20.9	20.6	21.0	19.9	17.6	14.2	11.2		
	10%	DB	17.8	20.6	24.3	29.0	31.8	33.6	33.5	33.3	30.6	27.0	22.3	17.9		
		MCWB	9.0	10.9	13.2	15.1	18.4	21.0	20.4	20.9	19.8	17.1	13.4	10.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.4	15.5	18.2	20.2	23.1	24.4	23.5	23.2	22.9	21.8	18.6	16.6		
		MCDB	18.1	20.7	23.5	27.2	30.7	31.8	31.9	31.7	29.9	27.0	22.3	19.7		
	2%	WB	12.3	14.4	16.9	19.1	22.0	23.5	22.8	22.6	22.1	20.6	17.5	14.6		
		MCDB	19.3	21.3	23.1	25.9	29.8	30.4	31.2	31.4	29.2	25.6	21.7	17.7		
	5%	WB	11.1	13.4	15.8	18.2	21.3	22.9	22.2	22.1	21.5	19.7	16.3	13.2		
		MCDB	18.3	20.2	21.9	24.8	28.8	29.8	30.7	31.0	28.5	25.1	21.3	18.5		
	10%	WB	10.0	12.3	14.7	17.3	20.5	22.2	21.7	21.6	21.0	18.6	15.1	11.4		
		MCDB	17.2	19.1	21.9	23.9	27.7	29.7	30.0	30.5	27.9	24.7	20.8	17.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.3	14.3	14.8	15.4	13.4	12.2	12.1	11.4	12.1	13.7	14.4	13.6		
		MCDBR	20.7	19.9	19.0	20.2	17.4	14.8	14.0	12.9	13.8	17.0	17.2	18.7		
		MCWBR	10.7	9.9	8.6	8.7	6.4	4.8	4.6	4.3	5.0	7.5	8.4	10.1		
		MCDBR	17.0	15.1	14.0	14.2	12.8	11.8	11.6	11.3	11.0	12.0	12.9	14.1		
	5% WB	MCWBR	9.4	8.1	7.4	6.8	5.5	4.4	4.0	3.9	4.2	5.8	7.2	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.273	0.289	0.324	0.350	0.371	0.387	0.364	0.353	0.307	0.285	0.266		
	taud		2.633	2.600	2.550	2.443	2.411	2.419	2.408	2.496	2.457	2.598	2.582	2.620		
	Ebn at Noon		989	1005	1001	967	937	912	895	917	919	951	953	970		
	Edh at Noon		77	88	99	114	119	117	118	107	107	87	81	75		
(44) All-Sky Solar Radiation	RadAvg		3.62	4.51	5.71	6.74	7.13	7.39	7.02	6.58	5.57	4.69	3.82	3.31		
	RadStd		0.36	0.47	0.34	0.45	0.42	0.35	0.44	0.36	0.37	0.49	0.40	0.29		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+92

Nomenclature: See separate page